

Carter Tull

513-805-6807 ▪ carterptull@gmail.com ▪ www.linkedin.com/in/cartertull

Software Engineer/Automation Developer

Software Engineering | Backend Development | AI Platform Development

Software engineer delivering production AI platforms and cloud-native automation solutions that transform complex content processing into scalable, reliable business systems within enterprise SaaS environments. Own technical initiatives from architectural design through production deployment and long-term optimization, combining software engineering, cloud infrastructure, and AI technologies to solve systems-level challenges across multiple services and repositories. Create reusable engineering foundations that strengthen software quality, accelerate development, and support long-term platform growth through shared architecture, automated validation, and scalable backend services. Contributions earned recognition from executive leadership for advancing enterprise automation initiatives, accelerating AI adoption, and delivering measurable value.

Core Competencies

- Platform Engineering
- Enterprise Automation
- Event-Driven Architecture
- Cross-Functional Collaboration
- REST API Development
- Cloud Architecture
- Google Cloud Platform (GCP)
- Large Language Model Integration
- Infrastructure as Code (Terraform)
- Software Development Lifecycle
- Application Architecture
- Microservices Architecture
- AI Evaluation Frameworks
- CI/CD Pipelines
- Cloud-Native Development

Professional Experience

CONSTRUCT CONNECT – Cincinnati, OH

June 2025 to Present

Construction technology SaaS provider delivering project intelligence, bid management, takeoff, and estimating software that helps contractors and manufacturers find, bid, and win construction work.

Software Engineer (April 2026-Present) | **Automation Developer** (June 2025-April 2026)

Engineer production AI platforms, cloud-native backend services, and Google Cloud Platform (GCP) solutions that automate construction project ingestion, enrichment, and publishing for Content Automation Services within Content division. Own end-to-end design, development, deployment, and lifecycle management of production microservices, AI services, and cloud infrastructure spanning development, QA, and production environments.

Develop Python, Java, TypeScript, SQL, and Terraform solutions that ingest construction documents and external partner data, orchestrate LLM-driven extraction workflows, and publish validated project information to enterprise content management systems. Architect scalable backend services, cloud-native application infrastructure, and deployment pipelines. Administer GitLab CI/CD, containerized workloads, cloud networking, IAM, Secret Manager, monitoring, and production operations. Coordinate cross-functional engineering initiatives spanning multiple repositories, services, and technology stacks.

- Reduced average bidding project publication time from 60 hours to 16.5 hours, a 73% improvement, which enabled automated publication of 50% of projects;** CEO recognized contributions as 1 of 6 key contributors to initiative.
- Built AI extraction evaluation framework from ground up;** established automated field, service, and project-level regression testing against golden datasets for model, prompt, and pipeline validation before production releases.
- Designed shared data layer for agentic AI platform, including schema architecture, version-controlled database migrations, OpenAPI contract, and automated CI drift enforcement.** Established system of record across services.
- Led migration of AI retrieval platform from Vertex AI to PostgreSQL with pgvector on Cloud SQL.** Resolved infrastructure and networking barriers, validated accuracy parity before production cutover, and achieved zero regression deployment.
- Replaced fuzzy string matching with an AI-powered comparator in contact matching service, which **increased successful contact matches by 12% while eliminating new false-positive matches.**
- Improved AI extraction accuracy from 82% to 85% and reduced end-to-end processing time 39%.** Refactored monolithic extraction architecture into parallel domain-specialized agents. **Increased accuracy to 87% and reducing latency further.**
- Built scheduled ETL pipeline** that retrieves construction opportunities from external partner APIs, transforms and de-duplicates records, and publishes validated projects through enterprise microservices.

- Directed migration of AI agent services from manually invoked cloud services to event-driven worker pools, which **eliminated manual execution, enabled resumable processing, and corrected data quality issue that allowed projects to publish under incorrect source company.**
- **Developed comprehensive onboarding documentation and created AI-powered onboarding assistant.** Delivered Git and GitHub training sessions for 100 to 125 employees and executives.
- Strengthened cloud security by **implementing least-privilege IAM, Secret Manager credential delivery, and encrypted private database connectivity.** Removed database passwords from Terraform state, remediated all high-severity dependency vulnerabilities, and restored production rollback capability across 3 environments.

THE OHIO STATE UNIVERSITY, CLASSROOM SERVICES – Columbus, OH

May 2023 to May 2025

Classroom Services Support Technician

FEDERAL HOME LOAN BANK OF CINCINNATI (FHLB) – Cincinnati, OH

May 2024 to August 2024

Software Engineering Intern

Education | Certifications | Professional Development

Bachelor of Science (BS), Computer & Information Science

The Ohio State University – Columbus, OH | 2025

- The Ohio State University Marching Band – 4-year scholarship member; 3-time Script Ohio i-dotter
- Member - Ohio State Competitive Programming Club

Anthropic Academy, February-April 2026

Completed advanced training in Claude, Claude Code, Anthropic API development, Model Context Protocol (MCP), Agent Skills, Subagents, Claude Cowork, and AI Fluency (Capabilities, Limitations, and Framework Foundations).

ConstructConnect AI Enablement Program, April 2026

Wave 1 Hackfest Participant (5-member team); selected among the Top 16 of 48 teams.

Microsoft Python Programming Fundamentals, Coursera, January 2026

Technical Skills

Programming Languages:	Python, TypeScript, JavaScript, Java, SQL, Terraform (HCL), C#, C, C++, Ruby
AI & Generative AI Technologies:	Anthropic Claude, Claude Code, Anthropic API, Model Context Protocol (MCP), OpenAI ChatGPT, OpenAI Platform Playground, Google Gemini, Vertex AI, GitHub Copilot, Glean AI
Cloud Platforms & Infrastructure:	Google Cloud Platform (Cloud Run, Cloud Run Services, Cloud Run Worker Pools, Cloud Functions v2, Cloud Run Jobs, Pub/Sub, Cloud Scheduler, Cloud SQL, Vertex AI, Secret Manager, IAM, Artifact Registry, Cloud Logging, Cloud Storage), Microsoft Power Platform (Power Automate, Power Apps), Amazon Web Services (AWS), Microsoft Azure
Application Development & DevOps:	Git, GitHub, GitLab, GitLab CI/CD, Docker, Terraform, Liquibase, Postman, Swagger/OpenAPI, REST API Design, Jira, Confluence, Miro
Frameworks & Server Technologies:	FastAPI, Flask, Spring Boot, Node.js, React, Next.js, Angular, NgRx, Rails, React Native, Expo
Databases:	PostgreSQL, pgvector, Cloud SQL, Firestore/Datastore, Microsoft SQL Server
Networking & Security:	REST, HTTP/HTTPS, OAuth 2.0, OpenID Connect (OIDC), TLS/SSL, VPC Networking, Private Service Connect
Development Tools & IDEs:	Visual Studio Code, PyCharm, IntelliJ IDEA, Cursor

Volunteering | Community Service

Eagle Scout, Boy Scouts of America – Troop 500 | 2012–2018

Patrol Leader, Troop Quartermaster, Troop Guide, and Troop Instructor